Unit 1 - Introduction to AI

Artificial Intelligence | Unit 1

What is Artificial Intelligence?

Artificial Intelligence (AI) is the simulation of human intelligence in machines that are programmed to think, learn, and solve problems. AI enables computers to perform tasks that typically require human intelligence such as visual perception, speech recognition, decision-making, and language understanding.

History of AI

- 1950 - Alan Turing proposes the Turing Test
- 1956 - Term "Artificial Intelligence" coined by John McCarthy
- 1960s - Early AI programs like ELIZA developed
- 1980s - Expert systems become popular in industry
- 1997 - IBM Deep Blue defeats chess world champion
- 2012 - Deep Learning revolution begins
- 2023 - Large Language Models like ChatGPT emerge

Types of AI

Narrow AI (Weak AI)

Designed for a specific task. Examples include Siri, Alexa, recommendation systems, and image recognition software.

General AI (Strong AI)

Hypothetical AI that can perform any intellectual task that a human can. Does not yet exist.

Super AI

AI that surpasses human intelligence in all areas. Theoretical concept for the future.

Applications of AI

- Healthcare - disease diagnosis, drug discovery
- Finance - fraud detection, algorithmic trading
- Transportation - self-driving cars, route optimization
- Education - personalized learning, intelligent tutoring
- Entertainment - content recommendation, game AI
- Manufacturing - quality control, predictive maintenance

AI vs Machine Learning vs Deep Learning

AI is the broad concept of machines performing smart tasks. Machine Learning is a subset of AI where machines learn from data. Deep Learning is a subset of ML using neural networks with many layers. Every Deep Learning system is ML, and every ML system is AI, but not vice versa.